

Bounded Perturbation Invariance for Deterministic Argmax

Research Architecture Runtime
feature/argmax

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Abstract

We study the finite deterministic argmax operator $\arg \max_{i \in [m]} s_i$ under additive score perturbations bounded in the ℓ_∞ norm. When a unique maximizer i^* exists with margin $\gamma(s) = s_{i^*} - \max_{j \neq i^*} s_j$, we prove that $\gamma(s) > 2\varepsilon$ is necessary and sufficient for i^* to remain the unique maximizer under every perturbation δ with $\|\delta\|_\infty \leq \varepsilon$. Verification in the repository is **computational** (ARGMAX_MARGIN_COMPUTATIONAL.V1), not a Lean/ITP formalization.

1 Introduction

Selection operators such as argmax are unstable near ties: infinitesimal score noise can change the reported index. A quantitative margin condition restores invariance under bounded noise. This note records the classical ℓ_∞ margin theorem as the first operator-specific result exercised end-to-end through the Discovery and Verification architectures in this repository.

2 Problem formulation

Fix $m \geq 2$ and scores $s \in \mathbb{R}^m$. Write $[m] = \{1, \dots, m\}$. The (set-valued) argmax is

$$\text{Argmax}(s) = \{i \in [m] : s_i = \max_{j \in [m]} s_j\}.$$

We write $i^* = \arg \max_i s_i$ when $|\text{Argmax}(s)| = 1$.

Perturbation model. For $\varepsilon \geq 0$, admissible perturbations satisfy $\|\delta\|_\infty = \max_i |\delta_i| \leq \varepsilon$. The perturbed scores are $s + \delta$.

3 Notation and definitions

Definition 1 (Margin). *If i^* is the unique maximizer of s , the margin is*

$$\gamma(s) = s_{i^*} - \max_{j \neq i^*} s_j.$$

Otherwise $\gamma(s)$ is undefined.

4 Mathematical assumptions

1. $m \geq 2$ and $s \in \mathbb{R}^m$.
2. A unique maximizer i^* exists, so $\gamma(s) > 0$.
3. Perturbations obey $\|\delta\|_\infty \leq \varepsilon$ with $\varepsilon \geq 0$.

We do not treat data-dependent scores $s_i(D)$, other ℓ_p balls, or set-valued selection rules beyond recording that ties exclude the hypothesis.

5 Main theorem

Theorem 2 (Bounded perturbation invariance). *Under the assumptions above, if $\gamma(s) > 2\varepsilon$, then for every admissible δ , i^* is the unique maximizer of $s + \delta$.*

Theorem 3 (Sharpness). *Under the same assumptions, if $\gamma(s) \leq 2\varepsilon$, then there exists an admissible δ such that i^* is not the unique maximizer of $s + \delta$.*

6 Supporting lemmas

Lemma 4 (Worst-case gap). *For any admissible δ and any $j \neq i^*$,*

$$(s + \delta)_{i^*} - (s + \delta)_j \geq s_{i^*} - s_j - 2\varepsilon \geq \gamma(s) - 2\varepsilon.$$

Proof. $(s + \delta)_{i^*} \geq s_{i^*} - \varepsilon$ and $(s + \delta)_j \leq s_j + \varepsilon$. Subtracting yields the first inequality; the second uses the definition of γ . $\square$

7 Proofs

Proof of Theorem 2. Lemma 4 and $\gamma(s) > 2\varepsilon$ imply $(s + \delta)_{i^*} > (s + \delta)_j$ for all $j \neq i^*$. $\square$

Proof of Theorem 3. Let $j^* \in \text{Argmax}_{j \neq i^*} s_j$. Set $\delta_{i^*} = -\varepsilon$, $\delta_{j^*} = +\varepsilon$, and $\delta_k = 0$ otherwise. Then $\|\delta\|_\infty \leq \varepsilon$ and

$$(s + \delta)_{i^*} - (s + \delta)_{j^*} = \gamma(s) - 2\varepsilon \leq 0,$$

so i^* is not a unique maximizer. $\square$

8 Discussion

8.1 Intuition

The margin is a buffer against adversarial noise that simultaneously depresses the winner and elevates a runner-up. Each direction can consume at most ε , hence the factor two.

8.2 Geometric explanation

8.3 Why the theorem matters

It gives an exact, checkable certificate that a reported argmax is stable to bounded score noise—useful for post-hoc selection and audit of ranking pipelines.

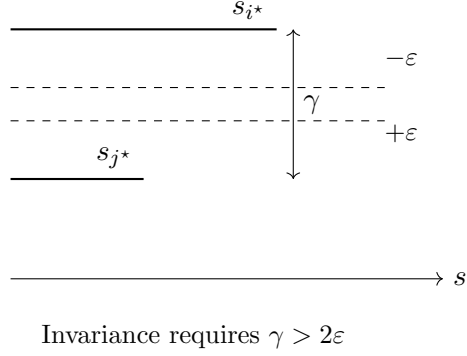


Figure 1: Margin buffer versus ℓ_∞ adversarial moves.

8.4 Where assumptions are used

Uniqueness supplies γ ; ℓ_∞ yields the additive 2ϵ shrinkage; finiteness of m is used only to ensure maxima exist. Indices in the paper are $[m] = \{1, \dots, m\}$; the reference implementation uses $0..m-1$ with the obvious shift.

8.5 Examples

Take $s = (3, 1, 0.5)$ so $\gamma = 2$. For $\epsilon = 0.9$, invariance holds. For $\epsilon = 1$, $\delta = (-1, +1, 0)$ yields $s + \delta = (2, 2, 0.5)$ (tie).

8.6 Counterexamples and boundary cases

At $\gamma = 2\epsilon$ the adversary forces a tie: uniqueness fails though i^* may remain a maximizer. Ties in s exclude the theorem hypothesis entirely.

9 Interpretation

The condition $\gamma > 2\epsilon$ is both a proof target and an operational test: compute the empirical margin and compare to the noise budget.

10 Utility implications

Stabilization mechanisms (score smoothing, hysteresis, top- k reporting) can be justified when the measured margin is too small relative to ϵ .

11 Limitations

- Verification in-repo is computational certification, not Lean/ITP.
- Finite score vectors; abstract s , not $s_i(D)$.
- ℓ_∞ only; other norms change constants.
- Floating-point evaluation can blur exact equality $\gamma = 2\epsilon$.

12 Open questions

Extensions to ties (set-valued `argmax`), ℓ_p balls, and data-dependent scores $s_i(D)$; composition with top- k and thresholding.

13 Connections to existing framework

Artifacts are minted as typed Discovery IR, packaged as a sealed CRP, and discharged by System B via `operators.argmax.verify` with evaluation method `ARGMAX_MARGIN_COMPUTATIONAL_V1`.

14 Verifier summary

End-to-end workflow reports audit verdict PASS with no unresolved obligations when the sealed claim statement exactly matches the canonical theorem text and formal fields. Feedback limitations include `COMPUTATIONAL_VERIFICATION_NOT_LEAN`.

15 Appendix

Canonical statement (repository string):

Let $m \geq 2$, $s \in \mathbb{R}^m$ with unique maximizer $i^* = \arg \max_i s_i$ and margin $\gamma(s) = s_{i^*} - \max_{j \neq i^*} s_j > 0$. Let $\varepsilon \geq 0$ and $\delta \in \mathbb{R}^m$ with $\|\delta\|_\infty \leq \varepsilon$. If $\gamma(s) > 2\varepsilon$, then i^* is the unique maximizer of $s + \delta$.

References

References

- [1] Repository implementation: `implementation/src/operators/argmax/` and
 `implementation/docs/operators/argmax.md`.